

Output Ontology (OO)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: MILU | Context: Mymensingh Environmental Scientist — Cross-Regional South Asian Agricultural Knowledge

Output Ontology definition: Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

MILU's output ontology is single-correct-answer MCQ over 41 India-derived subject labels and 8 domains [Q35, Q36, Q37, Q38]. The structural single-correct-label format cannot accommodate cross-border knowledge pluralism on contested topics like trans-border water policy — the deployment YAML identifies this as a HIGH-priority concern (transboundary_policy_context, GAP-3 [WEB-11, WEB-14]). Even if such questions existed, the format would force a single nationally-partial answer key (MAJOR Concern 7: DATASET-D5, D28). The 41-subject labels themselves were derived by clustering ~20K Indian-exam tags [Q35, Q36] and validated against Indian standards. Domain-wise model performance is poor in culturally relevant areas [Q16, Q68], and the authors note a need for 'more inclusive data distribution' [Q70]. For a MODERATE-priority dimension, the structural rigidity plus India-derived label space creates meaningful misalignment, though the basic MCQ correctness structure is functional for non-contested factual content.

ID	Evidence
Q16	"Our domain-wise analysis reveals that models perform poorly in culturally relevant areas, such as Arts & Humanities and Social Sciences, compared to more general fields like STEM." (p.2)
Q35	"Finally, in total, we get around 20K tags. However, these tags are highly fine-grained, often having a heavy overlap." (p.4)
Q36	"To organize them, we embed the tags using the NV-EMBED-V2 (Lee et al., 2024) model and apply K-means clustering to group tags into 50 clusters." (p.4)
Q37	"We manually review these clusters and assign appropriate subject labels." (p.4)
Q38	"Following the manual merging of related clusters, we determine 41 distinct subject names, which fall into eight main domains: Arts and Humanities, Social Sciences, Environmental Sciences, Law and Governance, Health and Medicine, Science, Engineering and Technology, and Business Studies." (p.4)
Q68	"Similar trends to those in Section (§5.2) are observed where the open models perform poorly in domains specific to Indian culture—such as Arts & Humanities, Social Sciences, and Law & Governance—but demonstrate higher performance in STEM fields." (p.7)
Q70	"Bridging this gap requires a more inclusive data distribution that ensures equitable representation of all cultures and languages." (p.7)
D5	নর্দ্বিবাচন কমশিনারক অপসারণ করা যতে পার... প্রধান নর্দ্বিবাচন কমশিনাররে সুপারশি রাষ্ট্রপত — Removal of Indian Election Commissioner — Indian constitutional law only
D28	राष्ट्रीय आपातकाल घोषित करने के लिए 'सशस्त्र वद्रोह' शब्द संवधान में कब जोड़ा गया?... 44वें संवधान संशोधन अधनियिम द्वारा — Indian constitutional amendment for 'armed rebellion' clause — Indian law and governance
WEB-11	Ganges treaty in renegotiation 2024–2026 — Indian and Bangladeshi positions diverge (https://thediplomat.com/2026/04/why-the-india-bangladesh-ganges-treaty-renewal-must-deliver-real-security/)
WEB-14	No formal Teesta treaty; positions categorically differ (https://en.wikipedia.org/wiki/Teesta_Water_Dispute)

Strengths

- The 41-subject taxonomy was derived through systematic clustering and manual review of ~20K fine-grained tags [Q35, Q36, Q37], reflecting principled construction rather than ad-hoc category assignment
- Domain-wise reporting allows users to disaggregate model performance by domain, enabling targeted diagnosis of weak areas like Arts & Humanities and Law & Governance [Q16, Q68]

- Cross-language structural consistency in label space supports comparative evaluation (DATASET Strength 4: D11)

ID	Evidence
Q16	"Our domain-wise analysis reveals that models perform poorly in culturally relevant areas, such as Arts & Humanities and Social Sciences, compared to more general fields like STEM." (p.2)
Q35	"Finally, in total, we get around 20K tags. However, these tags are highly fine-grained, often having a heavy overlap." (p.4)
Q36	"To organize them, we embed the tags using the NV-EMBED-V2 (Lee et al., 2024) model and apply K-means clustering to group tags into 50 clusters." (p.4)
Q37	"We manually review these clusters and assign appropriate subject labels." (p.4)
Q68	"Similar trends to those in Section (§5.2) are observed where the open models perform poorly in domains specific to Indian culture—such as Arts & Humanities, Social Sciences, and Law & Governance—but demonstrate higher performance in STEM fields." (p.7)
D11	ರಾಷ್ಟ್ರೀಯ ಆಹಾರ ಭದ್ರತಾ ಮಿಷನ್ ಒಂದು ಬೆಳೆ ಅಭಿವೃದ್ಧಿ ಯೋಜನೆ... ಮಣ್ಣಿನ ಆರೋಗ್ಯವನ್ನು ಪುನಃಸ್ಥಾಪಿಸಲು — National Food Security Mission crop scheme in Kannada — same Indian policy content replicated across languages

Checklist

Checklist item definitions:

ID	Assessment Question
OO-1	Review output label categories for regional relevance
OO-2	Identify missing regionally specific categories
OO-3	Flag categories encoding non-regional values
OO-4	Consider stakeholder-driven taxonomy redesign
OO-5	Document taxonomy issues harming structural/content validity

Checklist responses:

OO-1: Output labels are India-validated and India-derived [Q35, Q38]. Regional relevance for Bangladesh is partial: factual STEM labels generally transfer, but governance/law/policy labels reflect Indian constitutional and administrative categories not applicable to Bangladesh [DATASET-D5, D8, D28].

OO-2: Missing categories: no Bangladesh-specific agro-ecological label space (haor, beel, char-land), no trans-border hydrology label, no cross-border policy category that could accommodate plural answer keys. Confirmed absent [WEB-17, WEB-18].

OO-3: Indian state polity labels (Maharashtra Legislative Committee, Indian Election Commissioner removal) encode Indian-specific governance assumptions [DATASET-D5, D8, D17]. For environmental policy, Indian exam answers would encode India's official water-sharing positions, which differ from Bangladeshi authoritative positions on the Ganges and Teesta [WEB-11, WEB-14].

OO-4: Stakeholder-driven taxonomy redesign would be required to add (a) wetland ecology subdomains, (b) cross-border policy category supporting plural ground truths, and (c) technical agronomy depth subdomains (soil science, crop physiology, irrigation engineering).

OO-5: Documented taxonomy issues: India-derived label space [Q35, Q38], single-correct-answer structural rigidity (MAJOR Concern 7), and absence of regional categories. These violate structural validity (taxonomy doesn't fit cross-border pluralism) and content validity (missing categories) for the deployment.

ID	Evidence
Q16	"Our domain-wise analysis reveals that models perform poorly in culturally relevant areas, such as Arts & Humanities and Social Sciences, compared to more general fields like STEM." (p.2)
Q35	"Finally, in total, we get around 20K tags. However, these tags are highly fine-grained, often having a heavy overlap." (p.4)

Q38	“Following the manual merging of related clusters, we determine 41 distinct subject names, which fall into eight main domains: Arts and Humanities, Social Sciences, Environmental Sciences, Law and Governance, Health and Medicine, Science, Engineering and Technology, and Business Studies.” (p.4)
Q70	“Bridging this gap requires a more inclusive data distribution that ensures equitable representation of all cultures and languages.” (p.7)
D5	নর্বিচান কমিশনারকে অপসারণ করা যত্নে পার... প্রধান নর্বিচান কমিশনারের সুপারিশে রাষ্ট্রপতি — Removal of Indian Election Commissioner — Indian constitutional law only
D8	Which one of the following statements about Legislative Committee system of Maharashtra is not correct? — Maharashtra state governance — India-specific law, not relevant to Bangladesh
D17	కేరండి చుట్టిలో మహారాష్ట్రలో శాసనసభ కమిటీ వ్యవస్థ గురించి ఏ వరకటన సరైనది ఉంది? — Maharashtra Legislative Committee system in Telugu — state-specific Indian governance
D28	राष्ट्रीय आपातकाल घोषित करने के लिए 'सशस्त्र विद्रोह' शब्द संविधान में कब जोड़ा गया?... 44वें संविधान संशोधन अधिनियम द्वारा — Indian constitutional amendment for 'armed rebellion' clause — Indian law and governance
WEB-11	Ganges treaty in renegotiation 2024–2026 — Indian and Bangladeshi positions diverge (https://thedi diplomat.com/2026/04/why-the-india-bangladesh-ganges-treaty-renewal-must-deliver-real-security/)
WEB-14	No formal Teesta treaty; positions categorically differ (https://en.wikipedia.org/wiki/Teesta_Water_Dispute)
WEB-17	AI4Bharat/MILU GitHub: no Bangladeshi annotator participation (https://github.com/AI4Bharat/MILU)
WEB-18	NAACL paper: annotation by AI4Bharat volunteers and Indian exam-portal experts only (https://aclanthology.org/2025.naacl-long.507/)

Information Gaps

- Exact granularity of subject labels within Environmental Sciences (Agriculture, Geography, Earth Sciences, Environmental Science as separate sub-subjects?) — not directly observable beyond DATASET-D6, D7, D10

Requires Expert Verification

- A Bangladeshi policy expert's identification of which Indian exam answer keys would be considered authoritative vs. nationally partial for environmental governance questions

Remediation

Gap 1: Single-correct-answer MCQ structure cannot encode legitimate Indian–Bangladeshi pluralism on contested trans-border policy questions

Recommendation: For policy and trans-border items, adopt a multi-reference scoring schema where both Indian and Bangladeshi authoritative answers are accepted, or construct a separate pluralistic-ground-truth track using Bangladesh policy documents and the GBM bilateral context [WEB-11, WEB-14].

ID	Evidence
WEB-11	Ganges treaty in renegotiation 2024–2026 — Indian and Bangladeshi positions diverge (https://thedi diplomat.com/2026/04/why-the-india-bangladesh-ganges-treaty-renewal-must-deliver-real-security/)
WEB-14	No formal Teesta treaty; positions categorically differ (https://en.wikipedia.org/wiki/Teesta_Water_Dispute)